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NEURAL BELLMAN OPERATORS

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This paper develops Neural Bellman Operators for constructing feasible dynamic policies and bounding their loss against the original economic optimum. Own-future neural construction retains feasible action witnesses and explicit acquisition errors. Under recursive risk, reference-centered qualification gives state-dependent complete-policy comparisons. We derive an executable convex-order AVaR enclosure with second-order innovation resolution, and connect a complete admission-and-fallback policy to simultaneous economic resolving power. A new remotely frozen investment comparison matches design rank, labels and executable payload across ReLU, quadratic and adaptive spline continuations, while charging conventional fallback. The prespecified margin is one percent of the maximal first-date investment charge, with a simultaneous half-width budget below half that margin. The estimand uses a declared dyadic initial-state distribution and the original continuous innovation law. A separate acquisition-and-reuse region identifies which policy, work and price bounds justify deployment. Broader NBO applications and earlier adverse evidence retain their original hypotheses and records.

KEYWORDS: Dynamic programming, Neural approximation, Recursive risk, Policy qualification, Computational economics.

1. INTRODUCTION

A learned continuation is economically useful when it changes what can be computed and which decisions can be justified under the original law. Neural Bellman Operators (NBOs) connect a constructed continuation, its feasible action witnesses and the complete policy actually implemented. The central comparison is

$$\underline{V} \le V^* \le J^{\widehat{\pi}} \le \overline{J}.$$

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The lower inequality covers every original feasible action; the upper inequality includes observation, numerical action realization and the controller's subsequent choices. A guarantee for a fitted surrogate is not a substitute for this comparison.

Our first contribution is constructive. Each dated NBO is formed against its own already attained future, with explicit primitive state and action covers, integration allowances and feasible action owners. The affine–ReLU realization does not assume successful unrestricted nonconvex training. Original-owner compilation preserves the continuation and its feasible witness while changing evaluation work. Joint-domain and common-readout extensions retain the coupling between decisions at different acquired states. Their hypotheses and full statements remain available in the accompanying material.

Our second contribution is an original-law qualification account for complete policies. Under recursive risk, centering the original conditional dual at the stored continuation retains the penalty for changing the maximizing risk density. State-dependent subsolutions and supersolutions then compare the implemented policy with the original optimum. The continuous investment construction derives the required curvature, value supports and all-action inequalities from the original technology. Feasible risk witnesses minorize the lower action problem, and full-law envelopes dominate the policy value. These inequalities are valid for any supplied continuation satisfying their assumptions. Neural construction, representation quality, certificate validity and computational gain are therefore separate questions.

Our third contribution concerns economically useful precision. The preceding investment experiment established equivalence at a broad normalized tolerance; its intervals did not resolve small policy differences. We derive a convex-order AVaR enclosure with width proportional to the inverse square of innovation resolution, without requiring a differentiable value function or a fixed risk maximizer. Coupled with a complete admission-and-fallback policy, this gives a deterministic support for paired policy differences. The numerical and statistical budgets can consequently be chosen together before evaluation. The new margin is one percent of the largest first-date investment charge, eighty times smaller than the preceding margin, and the prospective simultaneous half-width budget is less than half that margin.

We test this account on a new independently generated initial-state stream. ReLU, projected-quadratic and adaptive spline continuations have the same six-dimensional design rank, labels and 128-byte executable payload. Their feature searches and all attempted fits are charged. The conventional online risk-DP comparator receives the same admission rule. Each fitted service retains rejected proposals and pays for its fallback; it is not relabeled as the unchanged R95 actor. The protocol and then the actual fitted models and calibration selector are committed before the confirmation stream is opened. All pairwise results, including unfavorable or unresolved comparisons, remain in the experiment.

The initial-state estimand is explicitly the mean under a uniform 20-bit dyadic state distribution. The state domain for construction and the subsequent innovation law remain continuous. Uniform residual qualification and pointwise first-date policy admission have different quantifiers. Repeated timing processes are not new economic observations. A separately coded rational implementation reconstructs stored proposals and original-law bounds on a random stratified audit, while independent exact arithmetic checks simultaneous inference on every paired row. Neither this audit nor the fixed-rank comparison is presented as a full independent learning-stack or maximal-vector replication.

The final link is a purchase decision. We separate one-time construction and audit charges from deployment work, and derive the region of workload counts and externally supplied computation prices for which a qualified service improves total cost. A policy-quality requirement remains an additional inequality. Fresh-process timings include startup, loading, checking, evaluation, fallback, output and shutdown, and their observed ranges support conditional sensitivity calculations rather than unmeasured guarantees on future machines.

Related work. Lipschitz extension and distance transforms provide established construction tools (McShane, 1934, Felzenszwalb and Huttenlocher, 2012). Approximate policy iteration, safe improvement and convex dynamic programming have established comparison results (Munos, 2003, Pirotta et al., 2013, Scherrer et al., 2015, Yang, 2020). The NBO construction retains the actual own-future approximation

and feasible action owner throughout its primitive accuracy account. A native and a neural encoding of the same function receive the same economic comparison.

Risk-averse Bellman equations and feature-based risk learning are established research directions (Ruszczyński, 2010, 2014, Ruszczyński and Yang, 2025). We do not claim elementary dual centering or the AVaR threshold identity as new. The new precision result combines convex quadrature order (Alomari, 2016) with that identity (Rockafellar and Uryasev, 2000), and connects its executable interval to complete-policy admission and prespecified resolving power. The concentration step is Hoeffding's inequality (Hoeffding, 1963), not a new concentration theorem. The comparison table in Section 8.5 locates the retained qualification results relative to their established mathematical tools.

Organization. The article presents own-future construction, primitive accuracy, complete-policy and recursive-risk qualification, continuous investment construction, decision-scale precision and the new frozen comparison. The Technical Supplement provides the complete proofs. The Numerical and Development Appendix retains the previous studies, joint-domain and deployment results, singular-transition analysis and adverse vector-scaling evidence as complete sections. Original diffusions, endogenous preferences, temporal selves and games retain their companion statements and assumptions. Historical content is preserved, not used as evidence that a different new empirical question has been answered.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$ and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the earlier expected-cost experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. Section 8 evaluates the same investment technology under an explicitly nested AVaR preference and retains the original continuous innovation law. The direct-cost inference in Section T is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It forms targets using its *own* fitted f_{t+1}, returns a neural continuation and a feasible policy, and records an economic error allowance. It does not receive the optimal continuation as a training label. Arbitrary trained networks can be evaluated by the first result below. The subsequent constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \leq \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No

transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^{\pi} f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^{\pi} - V_0^{*} \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 2.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \quad (5)$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in (5) have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \le f_t - \mathcal{T}_t f_{t+1} \le D_t k_t + e_t + 2L_t h_t, \quad (6)$$
$$\mathcal{T}_t^{\pi} f_{t+1} - f_t \le e_t + D_t \zeta_t,$$

and hence

$$J_0^{\pi} - V_0^* \le \sum_{t<T} B_t (D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T (2e_T + 2L_g h_T). \quad (7)$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 2.2.

The Lipschitz-extension principle is classical (McShane, 1934). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

3.1. *Original-owner compilation on a tensor cover*

On a tensor cover $\mathcal{X} = \prod_{j=1}^{d}\{x_{j,0},\dots,x_{j,n_j}\}$, put $S = \prod_j (n_j+1)$. For arbitrary labels y_i, including inconsistent inexact labels, compute at each grid node v

$$(z_v, o_v) = \min_{\text{lex,i}} (y_i + L\|v - x_i\|_1, i).$$

The index o_v is the original label owner, not the transformed node v.

THEOREM 3—Continuous original-owner identity: *For a closed tensor cell C with corners $\mathcal{V}(C)$, and $x \in C$,*

$$f(x) = \min_{v\in\mathcal{V}(C)} (z_v + L\|x - v\|_1), \tag{8}$$

$$(f(x), i_*(x)) = \min_{\text{lex},v\in\mathcal{V}(C)} (y_{o_v} + L\|x - x_{o_v}\|_1, o_v).$$

Coordinatewise forward and backward relaxations compute (z, o) in $O(dS)$ additions and comparisons and $O(S)$ storage. After locating the cell, a query uses $O(d2^d)$ arithmetic operations. Exact rational comparisons preserve all ties, including $L = 0$. Using the original owner's repaired action preserves the uncompiled policy, its cost and its certificate exactly.

The proof is in Supplement, Section 2.3.

The nodal algorithm is the classical distance transform (Felzenszwalb and Huttenlocher, 2012); preserving the original action through continuous off-grid queries is the policy obligation. If there are $J_{t,i}$ node actions and M_t integration representatives, write $Q_t = M_t \sum_i J_{t,i}$. Literal dense cone evaluation costs $O(\sum_t dS_{t+1}Q_t)$ in representation work. Compilation replaces this by $O(\sum_t [dS_{t+1} + Q_t d2^d])$, plus cell location, primitive evaluation, integration, feasible repair, certificate and output. Rational bit lengths and state boxes crossing many cells incur additional costs. There is no dimension-free complexity assertion.

3.2. *Acquired states and conventional actors*

A b-bit coordinate sensor reports a dyadic cell and its midpoint $\widehat{x}$. On the unit cube, $\|x - \widehat{x}\|_1 \le r(b) = d2^{-b-1}$. For the investment capacity $\kappa(x)$ below, clip the

original action to the smallest capacity throughout that cell and round down to an action quantum Δ_t. This is feasible at every state in the cell. Relative to an exact true-state repair, its displacement is at most $2K_t^A r(b) + \Delta_t$. Transporting the witness cone from $\widehat{x}$ to x adds at most $2L_t r(b)$, plus any certified selector objective error ν_t. Thus the implemented-policy account adds

$$\sum_{t<T} B_t\{(2L_t + 2D_t K_t^A)r(b) + D_t\Delta_t + \nu_t\}. \tag{9}$$

No continuity of the discrete owner selector is assumed. At uncertain numerical comparisons the simulator retains all possible actors, rather than choosing a favorable one.

For conventional FVI the continuation is multilinear on its realized tensor grid. Its actual modulus is bounded by the largest coordinate edge slope. A nearest-node actor uses a feasible node minimizer and true-state repair. Let h_t be the largest half-cell ℓ^1 radius, L_t^{gr} the Bellman graph modulus, and e_t the node query allowance. Its optimal residual upper bound is $D_t k_t + e_t + L_t^{\mathrm{gr}} h_t$. Its selected-policy upper bound is $e_t + E_t$, where

$$E_t = \sup_x \{y_{i(x)} + L_t^{\mathrm{gr}} \|x - x_{i(x)}\|_1 - f_t(x)\}.$$

Partitioning cells by nearest-node boundaries makes the expression separately affine in each coordinate. Its maximum is therefore at a partition vertex; the implementation encloses all such vertices. Both uniform and curvature-refined FVI receive this sharper one-sided account, not an artificially doubled baseline. Their acquired-state allowance is derived separately in (59).

4. ORIGINAL-OPTIMUM ACCURACY IN THE INVESTMENT ECONOMY

The exact witness theorem answers an implementation question conditional on a continuation. An economist also needs to know how close the complete implemented policy is to the original optimum. We now establish that connection in the investment economy itself. The argument exploits a property not used by the earlier cell-range calculation: despite the bilinear transition, the Bellman objective remains jointly convex. This permits a second-order interpolation account and transfers

verified nodal decisions to an implementable policy over the entire state domain. It does not assume that a fitted neural continuation is convex.

Relation to the preceding two-control query. The earlier query experiment used the fixed shared capacity $1/8$. The complete dynamic extension below uses the original state-dependent capacity $c(x)$, with the retained exposure columns, action-cost interaction and second shock. Its nesting claim concerns the original scalar economic model; it is not an assertion that a full dynamic policy was already present in the earlier isolated query.

4.1. *Primitives and Bellman regularity*

Let $d \geq 2$ be even, let indices be cyclic, and set $\beta = 15/16$. Write

$$S(x) = \frac{2}{d}\sum_j (x_j - 5/8)^2 + \frac{1}{4d}\sum_j (x_j - x_{j+1})^2 + 2\left(\frac{1}{2} - \frac{2}{d}\sum_j x_j\right)_+^2,$$

$$g(x) = S(x) + \frac{2}{d}\sum_j (x_j - 5/8)^2, \qquad c(x) = \frac{1}{8} + \frac{1}{8d}\sum_j x_j.$$

The original scalar control has $0 \leq a \leq c(x)$, running cost $S(x) + a^2 + 4a^4$, and transition

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + B_j a + s_j z,$$

where $(B_j, s_j) = (1/2, 1)$ at alternating coordinates and $(1/4, -1)$ at the others, and z is uniform on $[-1/32, 1/32]$.

For the retained two-control extension, $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the second exposure column interchanges $1/2$ and $1/4$, and the action cost is

$$R(a) = a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2/4.$$

An independent common-direction uniform innovation of radius $1/64$ is added. This is the previously specified extension, nested at $a_2 = 0$ when that innovation is disabled. Both models preserve $[0,1]^d$. Let $m \in \{1, 2\}$ denote the number of controls

and define

$$Q_t^*(x,a) = S(x) + R(a) + \beta \mathbb{E} V_{t+1}^*(F(x,a,Z)),$$
$$V_t^*(x) = \min_{a \in A(x)} Q_t^*(x,a), \qquad V_T^* = g.$$

All costs and laws in this display are the economic primitives, not a fitted surrogate.

THEOREM 4—Regularity of the original Bellman problem: *For either control specification and every finite horizon, V_t^* is convex and continuously differentiable in the interior, with the corresponding one-sided extension on the cube. It is G_t-Lipschitz in the ℓ^1 norm and M_t-semiconcave in the Euclidean norm, where valid constants are*

$$G_T = 10/d, \qquad G_t = \frac{1037}{128d} + \beta \frac{47}{64} G_{t+1},$$
$$M_T = 26/d, \qquad M_t = \frac{22}{d} + \frac{21}{256d} + \beta \left(\frac{9}{16} M_{t+1} + \frac{1}{8} G_{t+1} \right).$$

In particular, $G_t \le 27/d$ and $M_t \le 54/d$. The function Q_t^* is jointly strongly convex on its convex feasible graph, with state curvature at least $107/(128d)$ and action curvature at least $7/4$.

A valid semiconcavity constant along each coordinate line is

$$D_T = \frac{9}{d} + \frac{16}{d^2},$$
$$D_t = \frac{5}{d} + \frac{16}{d^2} + \frac{21}{256d^2} + \beta M_{t+1} \left(\frac{85}{256} + \frac{23}{256d} \right).$$

Consequently, for the positive multilinear interpolant I_n on a mesh of spacing $1/n$,

$$0 \le I_n V_t^*(x) - V_t^*(x) \le \kappa_t, \qquad \kappa_t = \frac{d D_t}{8n^2}. \tag{10}$$

The proof is in Supplement, Section 2.4.

The theorem is a structural statement about these economic primitives. It is not a claim that every controlled diffusion, recursive preference specification, or game in the complete development edition has a convex Bellman problem. Nor does it require a signed-weight neural fit to preserve convexity. Convex approximation of dynamic programs has an established literature; the additional work here is to

verify the needed curvature for the original nonlinear capacity-constrained model and connect it to the actual NBO actor and complete error account.

4.2. *Certified nodal Bellman bounds*

Suppose nodal arrays satisfy $l_{t+1} \le V^*_{t+1} \le u_{t+1}$ at all grid vertices. Equation (10) gives the functions $I_n l_{t+1} - \kappa_{t+1}$ and $I_n u_{t+1}$ as valid lower and upper continuations. The lower function need not itself be convex; convexity is used for the unknown true value and for the comparison argument, not asserted of arbitrary endpoint arrays.

Divide each uniform innovation component into q equal bins and use its conditional mean. Convexity gives a lower quadrature bound for the true continuation. Semiconcavity gives an upper error

$$\zeta_{t+1} = \frac{M_{t+1}}{6q^2} \sum_{r=1}^{s} \rho_r^2 \|s_r\|_2^2, \tag{11}$$

where $s = 1$ or 2, $\rho_1 = 1/32$, and $\rho_2 = 1/64$ in the extension. The expectation remains the original continuous law. At the terminal date g is evaluated directly, so no terminal interpolation allowance is needed.

At a vertex v, let $\mathcal{A}_A(v)$ be the capacity simplex lattice with spacing $c(v)/A$, including all its faces. For $m = 1, 2$, a minimizer can be represented by neighboring lattice vertices whose coordinate variances sum to at most $mc(v)^2/(4A^2)$. A valid action smoothness constant is

$$H_t = \frac{21}{4} + \beta M_{t+1} \|B\|_2^2, \quad \|B\|_2^2 = \frac{5d}{32}\ (m = 1), \quad \|B\|_2^2 = \frac{9d}{32}\ (m = 2).$$

Taylor's inequality and that barycentric representation imply

$$0 \le \min_{a \in \mathcal{A}_A(v)} Q^*_t(v, a) - V^*_t(v) \le \delta_t := \frac{mH_t}{128A^2}. \tag{12}$$

The linear Taylor terms cancel even when the optimum lies on a face. This is an error against the continuous feasible optimum, not only against the finite action menu.

Let $q_t^-(v,a)$ and $q_t^+(v,a)$ include all interpolation, quadrature and arithmetic errors and enclose $Q_t^*(v,a)$. Then

$$l_t(v) = \max\{0, \min_{a\in\mathcal{A}_A(v)} q_t^-(v,a) - \delta_t\}, \qquad u_t(v) = \min_{a\in\mathcal{A}_A(v)} q_t^+(v,a) \tag{13}$$

are valid nodal Bellman bounds. The zero truncation uses the nonnegative economic costs. Extra neural or conventional proposals may improve the upper side but do not remove the lower action-cover allowance. Positive interpolation, averaging and minimum maps are monotone and nonexpansive; the separate allowances in (10)–(12) are therefore propagated, rather than being discarded because an endpoint calculation contains the truth.

4.3. *From a fitted witness to the complete implemented policy*

Let $a_t(v) \in A(v)$ be the actual action returned at each vertex by any generator. It may be selected by the exact neural witness, a conventional fitted continuation, or their common verified fallback. Write $w_v(x)$ for the multilinear weights and set $\bar{\pi}_t(x) = \sum_v w_v(x) a_t(v)$. Affineness of capacity makes this action feasible. Let $q_t^{\mathrm{sel},+}(v)$ be a verified upper bound on its original-optimum continuation value $Q_t^*(v, a_t(v))$, using the same lower and upper Bellman reference for all generators.

THEOREM 5—Actual-policy, whole-domain certificate: *Set*

$$e_t = \max_v \{q_t^{\mathrm{sel},+}(v) - l_t(v)\} + \kappa_t.$$

Suppose the implemented policy $\widehat{\pi}_t$ is feasible and differs from $\bar{\pi}_t$ by at most b_t in action ℓ^1 norm, uniformly over the state cube. Define

$$\Lambda_t = \frac{13}{16} + \beta\frac{3d}{8}G_{t+1}, \qquad D_T^\pi = 0, \quad D_t^\pi = e_t + \Lambda_t b_t + \beta D_{t+1}^\pi.$$

Then, at every true state and date,

$$0 \le J_t^{\widehat{\pi}}(x) - V_t^*(x) \le D_t^\pi.$$

This statement is conditional on the actual recorded policy, not on success of its fitting algorithm. It applies equally to a pure fitted policy and to a policy augmented by verified common candidates.

The proof is in Supplement, Section 2.5.

In the execution, observation is rounded down at forty bits. An acquired state $y \leq x$ has no greater capacity than the true state. Each interpolated action is then rounded down at twenty bits using the lower endpoint of its arithmetic enclosure. If L_{ij} bounds the derivative of action coordinate i with respect to state coordinate j, and η bounds native interpolation error, a valid implementation allowance is

$$b_t \leq 2^{-40} \sum_{i,j} L_{ij} + m(2^{-20} + 2\eta + \eta_{\mathrm{out}}).$$

Here L_{ij} is computed from the maximum adjacent nodal difference times n, and η_{out} covers endpoint subtraction. All these numbers are recorded outward. This new barycentric acquired actor is explicitly distinguished from the earlier discontinuous cell-constant actor; an old actor is not silently replaced during a purported fixed-policy revalidation.

4.4. *Arithmetic, work, and scope*

The native interpolation kernel forms nonnegative tensor weights and a weighted sum of at most 2^d stored binary64 endpoints. It is compiled without reassociation or floating-point contraction. For $K = (3d+4)2^d + 32$, an explicit conservative absolute allowance is

$$\eta = \frac{K2^{-53}}{1 - K2^{-53}} \max_v |z_v| + K2^{-1022}.$$

The final term conservatively covers underflow. Input coordinates, affine indexing, finite endpoints and the binary64 format are checked. Roundoff expansion and the ordinary interval cost arithmetic remain separate from the analytic approximation bounds. Exact rational fixtures compare the native sum with its mathematical tensor interpolant.

With $N = (n+1)^d$ vertices, $K_A = A + 1$ scalar actions or $K_A = (A+1)(A+2)/2$ two-control actions, the reference calculation uses $O(TNK_A q^s 2^d d)$ elementary

work under the explicit tensor implementation. Stored reference values use $O(TN)$ words and a policy uses $O(TNm)$ words. Streaming batches avoid a state-pair table. Proposal fitting, exact rational witness work, actor transfer, serialization, independent cost inference and compilation are additional costs, not absorbed into this bound. The analytic errors are second order in state, action and innovation resolution; the state-count and tensor-evaluation factors still depend exponentially on dimension. The result supplies a matched original-optimum comparison without asserting that these costs have disappeared.

5. COMPLETE-POLICY QUALIFICATION AND ENDOGENOUS OCCUPATION

The first-date result holds the continuation policy fixed. A complete policy changes both later actions and the states at which they are taken. We now retain both changes. This extension concerns the original transition kernels on the continuous state space, rather than a replacement finite-state economy.

Fix a feasible reference policy τ and its bounded original-law values V_t^τ. In the expectation model of Section 2, define

$$A_t^\tau(x,a) = c_t(x,a) + \beta_t P_t^a V_{t+1}^\tau(x) - V_t^\tau(x). \tag{14}$$

Here a can denote a whole-cell feasible action rule, including acquisition and repair. At every date let $\mathcal{C}_t$ be a finite measurable acquisition partition. A menu label $a \in \mathcal{A}_{tC}$ denotes a measurable rule $x \mapsto a(x) \in A_t(x)$ for all $x \in C$. A parameter vector $\theta \in \Theta$ specifies the menu choice in every cell at every date; ties follow the declared measurable convention. The vector may share coordinates across dates and regimes. Let $\Theta_\epsilon = \{\theta \in \Theta : F(\theta) \le U + \epsilon\}$, where $U = F(\widehat{\theta})$ for a feasible attained vector. The qualification functional may use the candidate's own future. Convexity is not required for the transfer; a claimed exclusion must nevertheless have a valid certificate for this actual functional.

Write $v = (t, C, a)$ for a menu event. Retain any family $\mathcal{H}_\epsilon$ of finite event sets for which a verified calculation proves

$$\Theta_\epsilon \cap \bigcap_{v \in K} \{\theta : \pi_\theta \text{ selects } v\} = \varnothing, \qquad K \in \mathcal{H}_\epsilon. \tag{15}$$

A numerical infeasibility status alone is not such a certificate. The full-menu support duals in Section J give one implementation when labels and features yield the stated linear constraints. With nonlinear own-future dependence, those particular duals cannot simply be reused; verified outer relaxations or direct interval exclusion are required. Only strict separation excludes an event, so threshold equality remains included.

For every menu event and successor cell D, obtain uniform original-law enclosures

$$\begin{aligned} \ell_{tCa} \le A_t^\tau(x, a(x)) \le u_{tCa}, \qquad & x \in C, \\ \underline{p}_{tCaD} \le P_t(D \mid x, a(x)) \le \overline{p}_{tCaD}, \qquad & x \in C. \end{aligned} \tag{16}$$

The bounds must cover acquisition boundaries, feasible repairs and numerical error. Initial cell probabilities μ_C are known here; certified initial-probability constraints can instead be included in the feasible set. Averages under the reference policy are not substitutes for the uniform bounds in (16).

Define $\mathcal{P}_\epsilon$ by nonnegative occupation variables m_{tC}, q_{tCa}, transition flows z_{tCaD}, and $0 \le y_{tCa} \le 1$, subject to

$$\begin{aligned} & m_{0C} = \mu_C, \qquad \sum_a q_{tCa} = m_{tC}, \qquad \sum_a y_{tCa} = 1, \\ & q_{tCa} \le y_{tCa}, \qquad \sum_D z_{tCaD} = q_{tCa}, \qquad m_{t+1,D} = \sum_{C,a} z_{tCaD}, \\ & \underline{p}_{tCaD} q_{tCa} \le z_{tCaD} \le \overline{p}_{tCaD} q_{tCa}, \\ & \sum_{v \in K} y_v \le |K| - 1 \qquad (K \in \mathcal{H}_\epsilon). \end{aligned} \tag{17}$$

Every μ_C is nonnegative and $\sum_C \mu_C = 1$. Thus each m_t is a probability vector. These are outer occupation constraints, not a claim that the acquired cell process is Markov. In particular, they allow the within-cell state law to depend on all earlier actions.

THEOREM 6—Recursive common-parameter cost transfer: *Under the preceding boundedness, feasibility and enclosure conditions, put*

$$L_\epsilon = \min_{\mathcal{P}_\epsilon} \sum_{t,C,a} B_t \ell_{tCa} q_{tCa}, \qquad U_\epsilon = \max_{\mathcal{P}_\epsilon} \sum_{t,C,a} B_t u_{tCa} q_{tCa}. \tag{18}$$

Every $\theta \in \Theta_\epsilon$ satisfies

$$L_\epsilon \le J_0^{\pi_\theta} - J_0^{\tau} \le U_\epsilon.$$

The conclusion allows changes in every date's action and continuation, together with their induced original-law occupations. Valid additional compatibility or transition constraints can only tighten the enclosure. If ϵ-dependent exclusions are retained monotonically, the enclosures are nested as qualification is strengthened. Certified dual lower and upper bounds suffice; an optimizer's reported optimum does not.

For two complete-policy classes with reference-cost interval $[l_{jk}, u_{jk}]$ on a simultaneous event, their cost difference is enclosed on the same event by

$$[\, l_{jk} + L_j - U_k,\ u_{jk} + U_j - L_k \,].$$

No probability union over the parameter vectors is added after that event. A qualified bound on a reference policy's loss against V_0^* can also be enlarged by U_ϵ, retaining the original optimum as the benchmark.

The proof in Supplement 8 uses a telescoping identity under the candidate policy's occupation law, then embeds each complete candidate in (17). Independent choices at different cells and dates are not treated as jointly attainable. As in the first-date assignment bound, the relaxation itself can have unattainable points. The economic conclusion is an enclosure, not a worst-policy realization claim.

5.1. *Nonlinear recursion, temporal selves and games*

For a monetary evaluation ρ_t satisfying the monotonicity and cash-invariance conditions in Section 2, let $\mathcal{T}_t^\pi$ be the fixed-policy operator. Suppose, uniformly over the states and policies in the declared class,

$$d_t^- \le \mathcal{T}_t^\pi V_{t+1}^\tau - V_t^\tau \le d_t^+.$$

There is no linear-occupation identity for an arbitrary nonlinear ρ_t. Its correct counterpart is the following recursive order statement.

PROPOSITION 7—Recursive order and deviation bounds: *If* $e_T^- = e_T^+ = 0$ *and* $e_t^\pm = d_t^\pm + \beta_t e_{t+1}^\pm$, *then* $e_t^- \le V_t^\pi - V_t^\tau \le e_t^+$ *for every class member. More generally, on invariant recursive-value domains, if a monotone fixed-policy aggregator is* λ_t*-Lipschitz in its continuation argument and its reference residual is at most* b_t *in absolute value, then*

$$\|V_0^\pi - V_0^\tau\|_\infty \le \sum_{t<T} \left(\prod_{j<t} \lambda_j \right) b_t.$$

For a dynamic game, apply the one-sided sandwich of Proposition 1 to each player's best-response problem with the other players' complete strategies fixed. If that account is at most r_{it} *in every subgame, the attained profile has unilateral cost-improvement incentive at most* r_{it} *there. This is an approximate equilibrium certificate, not an equilibrium-existence assertion. Date-specific temporal selves are treated as distinct decision makers, with their stated objectives and future strategies, rather than as one artificially time-consistent planner.*

Endogenous preference states belong in the augmented state and transition before these hypotheses are checked. A diffusion approximation contributes its certified one-step operator error at every date, propagated with the same continuation moduli; an unproved weak-discretization rate is not assumed. The broader economic applications and their original hypotheses remain intact. The expectation identity, nonlinear order bound and strategic deviation certificate are distinct results, not interchangeable interpretations of one simulation table.

6. SIGNED TRANSITION QUALIFICATION FOR RECURSIVE PREFERENCES

The transport account can use more information than an absolute error radius. A numerical transition may have a known direction of bias, and the implemented action may have a more precise upper error bound than the whole feasible menu. We retain these distinctions. The resulting qualification applies to the original recursive evaluation at every date, not only to expected cost with a fixed continuation. It does not change an expectation objective into a risk objective or identify a static tail-risk criterion with its nested counterpart.

Retain the augmented economic state, original feasible actions and measurable complete policies of Section 2. Suppose bounded original values satisfy backward recursion with

$$Q_t(v;x,a) = c_t(x,a) + \beta_t \rho_{t,x,a}(v), \qquad T_t v(x) = \inf_{a \in A_t(x)} Q_t(v;x,a),$$

where each original conditional evaluation is monotone and cash invariant: $\rho(v + k) = \rho(v) + k$ for constants k. Discounts $\beta_t \geq 0$ are deterministic, $B_0 = 1$, and $B_t = \prod_{s<t} \beta_s$. The state includes the preference variables required by the original model. The relevant bounded functions are in the evaluation's domain, and the original Bellman optimum and supplied feasible policies are well defined. For each common parameter θ, store bounded $f_{t,\theta}$, a complete feasible π_θ, and numerical action values $\widehat{Q}_{t,\theta}$ on the original feasible actions.

Put $\widehat{T}_{t,\theta}(x) = \inf_a \widehat{Q}_{t,\theta}(x,a)$ and define the signed discrepancy

$$D_{t,\theta}(x,a) = Q_t(f_{t+1,\theta};x,a) - \widehat{Q}_{t,\theta}(x,a).$$

Assume uniform, verified bounds

$$\begin{aligned} \ell_{t,\theta} \leq \widehat{T}_{t,\theta}(x) - f_{t,\theta}(x) \leq u_{t,\theta}, \\ 0 \leq \widehat{Q}_{t,\theta}(x, \pi_{t,\theta}(x)) - \widehat{T}_{t,\theta}(x) \leq \gamma_{t,\theta}, \\ b^-_{t,\theta} \leq D_{t,\theta}(x,a) \quad (a \in A_t(x)), \\ D_{t,\theta}(x, \pi_{t,\theta}(x)) \leq b^{\pi,+}_{t,\theta}. \end{aligned} \tag{19}$$

Here $b^-_{t,\theta}$ and $b^{\pi,+}_{t,\theta}$ may be signed. A bound confined to sampled actions cannot replace the all-action lower bound. Original-owner repairs, unsampled feasible actions, integration error and arithmetic error must be included in the relevant inequalities. Let $d^-_\theta \leq g - f_{T,\theta} \leq d^+_\theta$ throughout the terminal domain.

THEOREM 8—Signed, recursive original-optimum qualification: *Under the stated conditions, every supplied complete policy satisfies, at every initial state,*

$$0 \leq V_0^{\pi_\theta} - V_0^* \leq \mathcal{S}_\theta,$$

$$\mathcal{S}_\theta = \sum_{t<T} B_t \{u_{t,\theta} - \ell_{t,\theta} + \gamma_{t,\theta} + b^{\pi,+}_{t,\theta} - b^{-}_{t,\theta}\} + B_T(d^+_\theta - d^-_\theta). \tag{20}$$

Consequently a class for which these inequalities hold uniformly and $\sup_\theta \mathcal{S}_\theta \le \varepsilon$ *consists entirely of complete policies with original-recursion loss at most* ε*. Actions, attained continuations and endogenous occupations may change at every date. The assertion is pointwise in the augmented state and does not require independent deployment contexts or a continuous actor.*

The bound is invariant to arbitrary date-specific, state-independent shifts of the stored continuations when the numerical action values are shifted consistently. If the original-minus-numerical action value is a constant $k_{t,\theta}$ *independent of state and action, its contribution to* (20) *is zero. A bias common to all actions at a given state, but varying across states, does not in general cancel. The coefficient one on* $b^{\pi,+} - b^-$ *cannot be reduced uniformly, even for one decision date and deterministic transitions.*

The proof compares both the complete policy and the original optimum with the same stored continuation by recursive order. It does not propagate a linear occupation identity through nonlinear preferences. Setting $b^- = -b$ and $b^{\pi,+} = b$ recovers Theorem 17; a verified signed enclosure can strictly improve it without a new fit or economic observation. With stationary bounded data and $0 \le \beta < 1$, the corresponding stationary bound is

$$V^\pi - V^* \le \frac{u - \ell + \gamma + b^{\pi,+} - b^-}{1 - \beta}.$$

The proof and the necessary bias counterexamples are in Supplement 15.

6.1. *Explicit qualification of two nonlinear evaluations*

Wasserstein distance controls expectations of Lipschitz functions. For nonlinear evaluations its price must be derived, not borrowed from the expectation model. For a bounded K-Lipschitz continuation f and probability laws $P, \widehat{P}$, define the

upper-tail average value at risk and entropic evaluation by

$$\rho_P^{\alpha}(f) = \inf_{z\in\mathbb{R}} \left\{ z + \frac{P(f-z)_+}{1-\alpha} \right\}, \qquad 0 \le \alpha < 1,$$
$$\rho_P^{\lambda}(f) = \lambda^{-1} \log P \exp(\lambda f), \qquad \lambda > 0. \tag{21}$$

The limit at $\lambda = 0$ is expectation. The first definition includes atoms; it is not the conditional mean obtained by discarding all mass at a quantile. The variational representation and its treatment of general loss distributions are established results (Rockafellar and Uryasev, 2002). We use these evaluations only when they are the original one-step primitives of the recursive economic objective.

PROPOSITION 9—Risk-specific transport allowances: *If* $W_1(P, \widehat{P}) \le e$, *then*

$$|\rho_P^{\alpha}(f) - \rho_{\widehat{P}}^{\alpha}(f)| \le \frac{Ke}{1-\alpha},$$
$$|\rho_P^{\lambda}(f) - \rho_{\widehat{P}}^{\lambda}(f)| \le K \exp\{\lambda \operatorname{osc}(f)\}\, e. \tag{22}$$

The entropic expression is invariant to constants added to f. *The factor* $(1-\alpha)^{-1}$ *is necessary uniformly over bounded Lipschitz continuations and probability laws. If a coupling satisfies* $f(Y) \ge f(\widehat{Y})$ *almost surely, the original-minus-numerical risk difference is nonnegative for either evaluation.*

If the additional numerical action-value error is at most ξ_t in absolute value, a symmetric choice in (19) is

$$b_t^{-} = -\beta_t \kappa_t e_t - \xi_t, \qquad b_t^{\pi,+} = \beta_t \kappa_t e_t + \xi_t,$$

where $\kappa_t = K_{t+1}/(1-\bar{\alpha}_t)$ for a verified risk-level cap $\alpha \le \bar{\alpha}_t < 1$, or $\kappa_t = K_{t+1}\exp(\bar{\lambda}_t O_{t+1})$ for verified bounds on the entropic parameter and continuation oscillation. A monotone coupling instead permits $b_t^{-} = -\xi_t$ and the same upper endpoint. Approximate evaluation of the exponential requires an outward upper bound; machine output is not an exact modulus.

These local risk coefficients enter the dated budget additively:

$$\sum_{t<T} B_t \eta_t + 2\sum_{t<T} B_{t+1}\kappa_t e_t + 2\sum_{t<T} B_t \xi_t + B_T(d^{+} - d^{-}) \le \varepsilon, \tag{23}$$

where η_t bounds residual span plus greediness. Monotonicity and cash invariance propagate constant error by β_t, not by a product of all later tail-risk multipliers. The continuation moduli themselves may grow with horizon or preference parameters and must still be established and paid for. Subject to this qualification, the coupled resource program of Section B allocates residual and risk-adjusted transition precision without deleting descendant or representation costs.

Economic and strategic interpretation. A larger tail-risk level can require finer transition information for the same original-policy loss allowance. That requirement is part of the original preference primitive, not a favorable reweighting of measured work. When preference levels evolve endogenously, the bounds must cover the augmented state and actual feasible actions. For games and temporal selves, the theorem applies to each fixed-opponent or fixed-future best-response recursion under that decision maker's original objective. Applying nested average value at risk to an originally static terminal-risk problem would change that problem; no such substitution is made. General non-cash-invariant recursive aggregators retain Proposition 7 and its separate continuation modulus.

7. REFERENCE-CENTERED RISK KERNELS AND COMPLETE-POLICY QUALIFICATION

A datewise worst-case error account can charge incompatible states at successive dates. Under nonlinear recursive preferences, replacing that account by an expectation under the physical transition law is not valid. We instead retain the conditional risk evaluation and center its dual representation at the actual stored NBO continuation. The resulting state-dependent certificate has an original-optimum interpretation and a finite occupation representation. It extends, rather than replaces, the signed qualification of Section 6.

7.1. *The centered operator*

Retain the original augmented economic state, feasible actions, nonnegative discounts, and bounded complete-policy recursions. At each (t, x, a) suppose the

original conditional evaluation admits the representation

$$\rho_{t,x,a}(v) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int v(y) q(dy) - \psi_{t,x,a}(q) \right\}. \tag{24}$$

Every q is a probability measure on the original successor state space. The representation is assumed on a common bounded-function domain containing all continuations below, with the measurability and selection conditions needed for the original Bellman recursions. This is a hypothesis about the original preference primitive, not permission to replace a static risk objective by a nested one. Conditional risk duality and risk-averse dynamic programming are established principles (Ruszczyński, 2010, 2014). For a nonconvex or non-cash-invariant aggregator lacking (24), the earlier order and modulus results remain applicable.

Fix one qualified parameter θ and suppress its subscript. Its entire sequence of stored continuations is $f = (f_t)_{t=0}^{T}$, and its supplied complete feasible policy is π. Define, using the same f_{t+1} in every term,

$$\begin{aligned} r_t(x,a) &= c_t(x,a) + \beta_t \rho_{t,x,a}(f_{t+1}) - f_t(x), \\ k_t^f(x,a,q) &= \psi_{t,x,a}(q) + \rho_{t,x,a}(f_{t+1}) - \int f_{t+1}\, dq, \\ \mathcal{E}_{t,x,a}^f(z) &= \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int z\, dq - k_t^f(x,a,q) \right\}. \end{aligned} \tag{25}$$

The penalty is nonnegative and has infimum zero, whether or not the supremum in (24) is attained. Direct substitution gives

$$\mathcal{E}_{t,x,a}^f(z) = \rho_{t,x,a}(f_{t+1} + z) - \rho_{t,x,a}(f_{t+1}). \tag{26}$$

In particular, $\mathcal{E}^f(0) = 0$, and $\mathcal{E}^f$ is monotone, cash invariant, and nonexpansive in the uniform norm. Discarding k^f is an upper relaxation, not this identity. Freezing one risk-maximizing kernel at f is in general not even an upper relaxation.

THEOREM 10—State-dependent original-optimum certificate: *Suppose verified functions satisfy $r_t^-(x,a) \le r_t(x,a)$ for every original feasible action and $r_t(x, \pi_t(x)) \le$*

$r_t^{\pi,+}(x)$. *Let bounded measurable* L_t, U_t *satisfy*

$$L_T \le g - f_T \le U_T,$$
$$L_t(x) \le \inf_{a \in A_t(x)} \{r_t^-(x,a) + \beta_t \mathcal{E}^f_{t,x,a}(L_{t+1})\},$$
$$U_t(x) \ge r_t^{\pi,+}(x) + \beta_t \mathcal{E}^f_{t,x,\pi_t(x)}(U_{t+1}). \tag{27}$$

Then, at every initial state,

$$f_0 + L_0 \le V_0^* \le V_0^\pi \le f_0 + U_0, \qquad 0 \le V_0^\pi - V_0^* \le U_0 - L_0. \tag{28}$$

The same statements hold at every intervening date and state. They are uniform over a qualified parameter class whenever all premises and the reported bound hold for each member with its own common continuation sequence and complete policy. The certificate permits endogenous states and actions changing at every date; it does not require a finite state space or independent deployment contexts.

With exact residuals and terminal discrepancies, equality recursions in (27) *give* $L_t = V_t^* - f_t$ *and* $U_t = V_t^\pi - f_t$. *With scalar bounds from* (19), *their corresponding constant lower and upper sequences are feasible certificates here. Thus the exact state-dependent recursion is no looser than the signed scalar account. A strictly smaller bound can result without changing a fit, a feasible action, or an economic observation.*

The last assertion concerns the best exact recursion or a verified improvement upon the constant certificates. It is not a claim that every approximately computed L, U improves the scalar account. In implementation, the minimum of valid policy supersolutions and the maximum of valid optimum subsolutions retain validity, so an inferior new bound need not displace the existing one.

The lower inequality requires every original action. A discrete action search without a verified original-menu lower bound cannot establish it. For a fixed action, a feasible risk kernel supplies a lower bound on $\mathcal{E}^f(L)$; an upper bound requires control of the entire risk envelope. Numerical integration, feasible repairs, residual errors and penalty errors belong in these inequalities. If penalty enclosures are used, lower penalty endpoints enter an upper certificate and upper endpoints enter

a lower witness. Approximate dual optimizers are accepted only through these independently checked inequalities.

7.2. *Exact risk occupations and safe kernel exclusion*

Consider now a finite-state implementation for which the original one-step risk has a finite affine representation

$$\rho_{t,s,a}(v) = \max_{j \in \mathcal{J}_{t,s,a}} \{q_{t,s,a,j}^{\mathsf{T}} v - \psi_{t,s,a,j}\}.$$

The supplied complete policy is fixed before the following maximization. Write $r_{ts} = r_t(s, \pi_t(s))$, $k_{tsj} = k_t^f(s, \pi_t(s), q_j)$ and $q_{tsj} = q_{t,s,\pi_t(s),j}$. For an initial probability vector μ, let $\mathcal{Y}^\pi$ consist of nonnegative masses satisfying

$$m_{0s} = \mu_s, \qquad \sum_j y_{tsj} = m_{ts}, \qquad m_{t+1,z} = \sum_{s,j} y_{tsj} q_{tsj,z}. \tag{29}$$

These are risk-distorted occupations, not frequencies under the physical law. Initial integration by μ is outside the conditional value function; a different original initial risk aggregation needs its own evaluation.

PROPOSITION 11—Penalized occupation identity and dual witness: *For the preceding finite representation,*

$$\mu^{\mathsf{T}}(V_0^\pi - f_0) = \max_{\mathcal{Y}^\pi} \Bigg\{ \sum_{t<T,s} B_t r_{ts} m_{ts} - \sum_{t<T,s,j} B_{t+1} k_{tsj} y_{tsj} + B_T \sum_s (g_s - f_{Ts}) m_{Ts} \Bigg\}. \tag{30}$$

Any feasible occupation gives a lower bound on this fixed-policy value. Any U with $U_T \geq g - f_T$ and

$$U_{ts} \geq r_{ts} + \beta_t(q_{tsj}^{\mathsf{T}} U_{t+1} - k_{tsj}) \quad \textit{for every } t, s, j$$

gives an upper bound $\mu^{\mathsf{T}} U_0$. Matching witnesses certify equality using rational arithmetic when the input is rational. The optimal-policy recursion retains the order $\inf_a \sup_q$; it is not obtained by minimizing jointly over policy actions and risk kernels.

A stored continuation can also exclude risk kernels, not just action events. Fix a feasible retained witness q_0 and a verified set $\mathcal{D}$ containing the successor error vector to be evaluated. Define its support function $h_{\mathcal{D}}(d) = \sup_{z\in\mathcal{D}} d^{\mathsf{T}} z$. Then the strict certificate

$$h_{\mathcal{D}}(q - q_0) < k^f(q) - k^f(q_0) \tag{31}$$

excludes q from the maximizing affine pieces throughout $\mathcal{D}$. Equality is retained. A verified upper support bound and lower bound on the penalty difference suffice. For a box $L \le z \le U$ the support is

$$\sum_i (q_i - q_{0i})_+ U_i + \sum_i \min\{q_i - q_{0i}, 0\} L_i.$$

For $\operatorname{osc}(z) \le s$, it is $s\|q - q_0\|_1/2$. The error set must be justified before exclusions are used; defining it from the same unverified pruned computation is circular. Unpruned signed or state-dependent certificates supply one valid starting set.

7.3. *Preference primitives and computational implications*

For average value at risk, $q_i \le p_i/(1-\alpha)$, $q_i \ge 0$ and $\sum_i q_i = 1$, with zero penalty on that capped simplex. Hence $k^f(q) = \rho_p^{\alpha}(f) - q^{\mathsf{T}} f$. Quantile atoms and zero-probability outcomes are retained (Rockafellar and Uryasev, 2002). For entropic evaluation, put

$$p_f(dy) = \frac{e^{\lambda f(y)} p(dy)}{\int e^{\lambda f}\, dp}.$$

Its centered penalty is $k^f(q) = \lambda^{-1}\mathrm{KL}(q\|p_f)$, and therefore

$$\mathcal{E}^f(z) = \lambda^{-1} \log \int e^{\lambda z}\, dp_f. \tag{32}$$

The change of measure is a computational identity, not a change to the original economic law or its risk parameter. Supplement 16 proves the representation and its support conditions. A finite affine occupation program is exact only when the stated finite representation is exact; discretizing the entropic dual without an enclosure does not meet that condition.

For a fixed finite representation, backward certificate evaluation uses at most a constant multiple of

$$\sum_{t,s,a} \sum_{j \in \mathcal{J}_{t,s,a}} |\operatorname{supp}(q_{t,s,a,j})|$$

arithmetic operations once residuals and penalties are available. It stores successive value layers rather than expanding descendant histories. Kernel construction, original-law integration, neural-circuit evaluation, support certificates, rational bit lengths and stored witnesses are additional costs. Risk-envelope enumeration can itself be large. This bound does not remove state-dimension dependence, certify an unverified continuous-state aggregation, or prove a speed advantage over another implementation.

For fixed penalties, increasing residuals by nonnegative local allowances e_{ts} increases the fixed-policy upper value by no more than

$$\max_{\mathcal{Y}^\pi} \sum_{t,s} B_t e_{ts} m_{ts}, \tag{33}$$

including a terminal allowance at $t = T$. This gives an occupation-sensitive sufficient precision budget in place of $\sum_t B_t \max_s e_{ts}$. The budget prices allowable numerical errors; an application-specific price of computation remains a separate input. Shared-parameter restrictions must be retained before taking an envelope over policies, as in Section K.

Interpretation. The new mechanism depends on the stored continuation through k^f and on the original preference through its risk envelope. Equivalent neural and non-neural representations have identical centered guarantees. A representation-specific computational benefit requires a matched execution, not a validity theorem. The existing held-out ReLU result, warning records, unresolved economic comparisons, and adverse vector horizon evidence therefore remain unchanged. The exact finite diagnostics below test the nonlinear certificate and its independent implementation; they are not a substitute for the original large-vector comparison or new economic observations.

8. CONTINUOUS-LAW QUALIFICATION IN THE INVESTMENT MODEL

The risk theorem becomes a numerical method only when its inequalities can be constructed from economic primitives. We now supply this construction for the original investment technology with a nested tail-risk preference. States and innovations remain continuous, every feasible action is covered, and the implemented terminal controller rather than a hypothetical exact optimizer determines policy cost. This application joins continuation construction, the original-optimum comparison and a representation comparison on one problem. It does not replace the general NBO construction or the other economic applications.

8.1. *Primitives and the object being approximated*

Let $d \geq 2$ be even, $X = [0,1]^d$, and interpret subscripts cyclically. At dates $t = 0, 1$, the feasible investment is $0 \leq a \leq A(x) = 1/8 + \bar{x}/8$, where $\bar{x} = d^{-1} \sum_j x_j$. Write $b_j = 1/2$ for odd j and $b_j = 1/4$ for even j, and $s_j = (-1)^{j+1}$. The original transition is

$$F_j(x, a, z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1 - x_{j+1})}{16} + b_j a + s_j z, \qquad z \sim \mathrm{Unif}[-1/32, 1/32]. \tag{34}$$

Define

$$S(x) = \frac{2}{d} \sum_j (x_j - 5/8)^2 + \frac{1}{4d} \sum_j (x_j - x_{j+1})^2 + 2(1/2 - 2\bar{x})_+^2, \tag{35}$$

$$g(x) = S(x) + \frac{2}{d} \sum_j (x_j - 5/8)^2, \qquad h(a) = a^2 + 4a^4.$$

We use $\beta = 15/16$ and, separately, $\alpha \in \{1/2, 3/4\}$. Average value at risk is the upper-tail cost functional

$$\rho_\alpha(v) = \sup \left\{ \int v q \, dP : 0 \leq q \leq (1 - \alpha)^{-1}, \int q \, dP = 1 \right\}.$$

Here P is the original uniform innovation law. The objective is recursively defined by $V_2 = g$ and

$$V_t(x) = S(x) + \inf_{0 \leq a \leq A(x)} \{ h(a) + \beta \rho_\alpha(V_{t+1}(F(x, a, \cdot))) \}.$$

It is not the static tail risk of total realized cost. A complete policy value replaces each infimum by the action prescribed by that policy, with the same recursive risk primitive.

Assumptions for the continuous construction. For this subsection, the primitives are exactly (34)–(35); α is fixed before construction; the fitted continuation f is a finite convex function on X; and every numerical lower or upper value is an outward enclosure. Feasible tail densities and feasible actions are retained explicitly. Terminal selection is the measurable, dyadically acquired controller described below, with its fallback included in the error bound. These concrete conditions imply compact feasible graphs, bounded costs, measurable minimizers, and the conditional-risk dual used in Theorem 10. They do not assert that every recursive preference or every trained network has this convex structure.

PROPOSITION 12—Primitive curvature and state supports: *For either risk level and every even d, the one-date objective $Q_1(x,a) = S(x) + h(a) + \beta\rho_\alpha(g(F(x,a,\cdot)))$ is jointly convex on $X \times [0,1/4]$, with curvature bounded below by*

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2 \tag{36}$$

in a direction (v,w). Its value V_1 is convex on X and satisfies

$$|V_1(x) - V_1(y)| \le \frac{15}{d}\|x - y\|_1. \tag{37}$$

Let W be the objective evaluated under any fixed feasible risk density, at an anchor (x_0,a_0), and let (p_x,p_a) be its gradient there. For any $\lambda,\nu \ge 0$, set $e = p_a + \lambda - \nu$. Then the following is a global affine minorant of V_1:

$$\ell(y) = W + \lambda(a_0 - A(x_0)) - \nu a_0 - \frac{e^2}{4} + \left(p_x - \frac{\lambda}{8d}\mathbf{1}\right)^\top (y - x_0). \tag{38}$$

Neither optimal multipliers nor an exact optimizing action are required.

The curvature is obtained from the original nonlinear technology, not imposed on an interpolated value grid. The negative second-order term from $x_j(1 - x_{j+1})$ is bounded using $|\partial_j g| \leq 10/d$; the stage curvature dominates it. The allowance $e^2/4$ pays for imperfect stationarity. The proof in the Technical Supplement also establishes the global state modulus in (37), including the state-dependent action constraint.

8.2. *An all-action certificate under the original innovation law*

Partition the innovation interval into n equal bins B_i, with midpoints z_i and probabilities $p_i = 1/n$. An innovation bin is a set over which an inequality holds, not a replacement point mass in the economic model. Choose nonnegative $q_i \leq p_i/(1-\alpha)$ summing to one. These masses define a feasible density constant on each bin. For a fixed initial state and feasible proposal a_0, let W_0 be a lower bound on the objective under this density, obtained from globally valid affine supports of the continuation at the successor midpoints. Let p enclose the derivative of the same supporting objective at a_0. Then

$$\underline{V}_0(x) = \inf_{0 \leq u \leq A(x)} \{W_0 + p(u - a_0) + (u - a_0)^2\} \tag{39}$$

is a lower bound on the original Bellman optimum when the supported continuation is V_1. The scalar quadratic is minimized over the full original interval. Interval uncertainty in p is covered by taking the smaller minimum at its two endpoints, since the minimum is concave in p.

For a convex fitted continuation, midpoint supports give the lower risk witness and the larger endpoint value bounds the continuation throughout each bin. For V_1, Proposition 12 supplies midpoint supports and the allowance $15/(32n)$ covers each half-bin. Upper tail risk is the average of the largest $n(1-\alpha)$ upper bin bounds. This average is the exact risk of a dominating step function, hence an upper bound under the continuous law.

THEOREM 13—Constructed original-optimum qualification: *Suppose the supports and bin bounds above are outwardly verified. Suppose further that the complete terminal controller satisfies $V_1 \leq V_1^{\widehat{\pi}} \leq V_1 + \varepsilon_1$ everywhere on X. For every queried initial state, the*

lower value (39) *and the binwise upper bound* $\overline{J}_0^{\widehat{\pi}}$ *satisfy*

$$0 \le J_0^{\widehat{\pi}}(x) - V_0(x) \le \overline{J}_0^{\widehat{\pi}}(x) - \underline{V}_0(x). \tag{40}$$

For any stored convex continuation f*, the same lower and upper constructions are admissible bounds for the centered-risk comparison of Theorem 10. Their validity does not depend on the representation of* f*, on a successful fit, or on the proposal being an exact minimizer.*

To see the connection, write a verified continuation bracket as $\ell \le V_1 \le u$ and use errors $L = \ell - f$, $U = u - f$. The centered identity gives

$$\rho_\alpha(f) + \mathcal{E}^f(L) = \rho_\alpha(\ell), \qquad \rho_\alpha(f) + \mathcal{E}^f(U) = \rho_\alpha(u).$$

Joint evaluation avoids subtracting and then adding two separately rounded copies of $\rho_\alpha(f)$. The bound (39) uses one feasible risk density to minorize the required all-action infimum; the policy upper bound controls the full risk envelope. Thus the application constructs sufficient inequalities for the original theorem. It does not claim that a non-neural continuation would receive a different validity theorem, or that computing the original-value witness is free.

The terminal controller observes each state coordinate rounded down to 40 binary places. It accepts its floating proposal only after an outward all-action gap of at most 2^{-21} at that observed state; otherwise it uses rational bisection. The original-state transfer is less than $30\,2^{-40}$. Consequently $\varepsilon_1 = 2^{-20}$ covers the complete controller, including observation and fallback. Policy evaluation encloses this controller at every possible successor, not only at the midpoint states used by the integrator.

8.3. *Residual covers and strict continuum screening*

A cellwise residual bracket need not solve another action problem at every state. On each rectangular state cell, convex interpolation of the vertex upper values dominates V_1; each affine function (38) minorizes it globally. The same interpolation and subgradient construction applies to convex f. Comparing an affine minorant of one function with the convex interpolation upper bound of the other at every vertex proves a bound for $V_1 - f$ throughout the cell. Taking the best available anchor and

then the worst cell gives a whole-domain residual interval. The 32×32 cover below therefore qualifies the continuous square, rather than converting its $1,089$ nodes into empirical generalization observations.

Let $[l_i, u_i]$ enclose an original successor value over every point of bin B_i. If τ is the kth largest l_i, $k = n(1-\alpha)$, then

$$u_i < \tau \tag{41}$$

excludes all risk mass from bin i in an upper-tail maximizer. At least $1-\alpha$ of the original probability mass has value at least τ; strict separation makes any positive mass on the excluded bin suboptimal. Equality is retained. These bounds must be available before screening. The screening calculation below pays for their construction and for the test itself, and compares identical upper-tail values with and without screening. This continuum-bin result is separate from enumerating all extreme kernels of a finite risk envelope.

8.4. *A fixed-accuracy horizon account for the original scalar and vector economies*

The continuous two-date implementation is not the limit of the constructive argument. The original scalar model and its retained two-control, two-innovation extension in Section 4 admit a horizon-uniform primitive account under nested AVaR. For the vector extension, actions satisfy $a_1, a_2 \geq 0$, $a_1 + a_2 \leq A(x)$; the second exposure column interchanges $1/2$ and $1/4$, the action cost is $a_1^2 + a_2^2 + 4(a_1^4 + a_2^4) + a_1 a_2/4$, and an independent common-direction shock of radius $1/64$ is added. These are the retained original vector primitives, not a new model selected to obtain a horizon bound.

PROPOSITION 14—Uniform risk regularity and constructive accuracy: *Consider either economy at any finite horizon, with $\beta = 15/16$ and a prespecified risk level $\alpha_t \in \{1/2, 3/4\}$ at each date. Then the original value functions are convex and $27/d$-Lipschitz in the ℓ^1 norm. Their Bellman objectives have joint state curvature at least $107/(128d)$ and action curvature at least $7/4$ on the feasible graph. No continuous differentiability of a risk-averse value function is asserted.*

The own-future cone construction of Theorem 2 may use the same modulus $27/d$ at every date. Let its state grid have n_x intervals per coordinate; let the action net have resolution m on the original capacity interval or capacity simplex; and let each innovation coordinate have n_z equal bins, where n_z is a multiple of four. With exact labels for the finite-bin risk calculation, the charged continuous-law integration allowance and exact feasible repairs, the constructed complete policy satisfies

$$J_0^{\pi} - V_0^* \le 16\left\{\frac{27}{n_x} + \frac{1319}{512m} + \frac{1215}{512n_z}\right\}. \tag{42}$$

The bound holds for every horizon with these same resolutions. Acquisition and final action-rounding allowances are added according to (9); reserving any positive share of the target tolerance for them gives finite sensor and action precisions independent of the horizon.

For an approximation allowance $\varepsilon_0 > 0$ before acquisition, sufficient choices are

$$n_x \ge \frac{1296}{\varepsilon_0}, \qquad m \ge \frac{3957}{32\varepsilon_0}, \qquad n_z \ge \frac{3645}{32\varepsilon_0}, \quad 4 \mid n_z.$$

These conservative sufficient resolutions are not a practical recommendation. They make explicit what was absent from a node-count statement alone: fixed original-optimum accuracy can be obtained with resolutions independent of the horizon in these original discounted economies.

Put $S = (n_x+1)^d$, let $A_m = m+1$ in the scalar case and $A_m = (m+1)(m+2)/2$ in the vector case, and let $s \in \{1,2\}$ be the innovation dimension. A dense stored implementation uses $O(TdS^2A_m n_z^s)$ rational arithmetic operations, plus risk sorting and output. Original-owner compilation replaces continuation evaluation by

$$O\Big(T\{dS + SA_m n_z^s\, d[2^d + \log(n_x+1)]\}\Big)$$

operations, plus the same primitive, risk-sorting and output charges. The state and innovation costs remain explicit. Fixed precision is not presumed: a common-denominator realization has rational bit length $O((T+1)\log D)$ for a fixed polynomial-size primitive denominator D. Thus operation counts are linear in the horizon at fixed accuracy and dimension, while bit-operation costs also pay for growing exact numbers. This constructive result does not assert that the largest

vector run has been independently executed, or that its conservative worst-case resolutions are economically competitive.

8.5. *What is established by each part of the method*

Markov risk dynamic programming, conditional-risk duality and policy iteration are established results (Ruszczyński, 2010, 2014). Feature-based risk evaluation and structured policy improvement are also established research directions (Ruszczyński and Yang, 2025). We do not claim these objects, or elementary dual centering, as new. Table I separates the prior mathematical tool from the conclusion obtained here.

Three levels must remain distinct. The bounded-function risk theorem is analytical and applies under its stated dual and selection assumptions. The finite affine-risk occupation result has an exact rational implementation. The continuous investment construction has outward numerical enclosures and primitive-specific convexity. Neither of the latter two implementations establishes numerical tractability for all instances of the first. The NBO contribution includes constructive continuations, their attained-policy qualification and their resource account; a validity theorem need not privilege a neural parameterization for that research question to remain meaningful.

9. DECISION-SCALE PRECISION AND THE COST OF A QUALIFIED NBO

An approximation can save continuation queries and still leave an economic comparison unresolved. The investment experiment in Section D illustrates this distinction: the original policy intervals establish equivalence only at a margin equal to four fifths of the maximal first-date investment charge. We now connect numerical integration, policy admission, statistical resolution and the cost of deployment. The original economic law and the NBO construction remain the benchmark. Neither a smaller numerical interval nor a new admission rule is silently attributed to the unchanged historical actor.

TABLE I

ESTABLISHED TOOLS AND THE ADDITIONAL QUALIFICATION OBJECT

Tool	Additional information retained	Conclusion in this paper
Risk Bellman recursion	Actual stored continuation and complete feasible controller	State-dependent bounds against the original optimum, not a surrogate objective.
Residual comparison	Signed local discrepancies and reference-centered dual penalties	A policy upper inequality and an all-original-action lower inequality; harmless offsets need not be charged twice.
Finite occupation duality	Risk-distorted flows and penalties attached to the same continuation	Exact finite comparison with the original $\inf_a \sup_q$ order.
Affine support domination	An independently verified error set	Strict safe exclusions; no exclusion at equality and no circular construction of the error set.
Convex risk optimization	Primitive curvature, imperfect multiplier residuals and continuous-law bin bounds	Computable whole-action lower witnesses and full-controller upper costs, Theorem 13.
Feature-based learning	Frozen labels, coefficients, selector and common original-optimum target	A reproducible comparison of representation, policy cost and verification work, not a presumption of neural superiority.

9.1. *Risk quadrature from the convexity of the original value*

Fix a current state and feasible action. Let $v(z) = V_1(F(x,a,z))$ on $[-R,R]$, where $R = 1/32$. Proposition 12 implies that v is convex and 15-Lipschitz. This conclusion concerns the true one-date value. The complete terminal controller need not be convex: its original bound $V_1 \le V_1^{\widehat{\pi}} \le V_1 + \varepsilon_1$, $\varepsilon_1 = 2^{-20}$, will be added separately.

Let $z_0, \dots, z_n$ be the endpoints of equal innovation bins, and let $m_i = (z_{i-1} + z_i)/2$. Define M_n to put probability $1/n$ at each midpoint. Define E_n to put probability $1/(2n)$ at each of the two endpoints of every bin, counting an interior endpoint

twice. These are numerical measures used for enclosures, not alternative economic transition laws.

PROPOSITION 15—Convex-order risk enclosure: *For a convex K-Lipschitz function v and the uniform law P on $[-R,R]$,*

$$\rho_\alpha^{M_n}(v) \le \rho_\alpha^{P}(v) \le \rho_\alpha^{E_n}(v), \qquad \rho_\alpha^{E_n}(v) - \rho_\alpha^{M_n}(v) \le \frac{KR}{(1-\alpha)n^2}. \tag{43}$$

If midpoint lower values and endpoint upper values have errors at most e_- and e_+ respectively, the computed width is at most the last term plus $e_- + e_+$. A terminal controller uniformly within ε_1 of V_1 adds only $\beta\varepsilon_1$ to its first-date upper cost. All numerical endpoint arithmetic must be enclosed separately.

The proof combines the established convex midpoint–trapezoid order with the AVaR threshold representation; neither underlying principle is claimed as new (Alomari, 2016, Rockafellar and Uryasev, 2000). Their use here changes the implementable original-law precision account. The previous full-bin Lipschitz allowance decays as n^{-1}. The new width decays as n^{-2} without assuming a differentiable risk-averse value or a fixed maximizing risk density. Both numerical risks are finite weighted tail calculations. Ignoring the half weights at the two outer endpoints would compute a different bound.

COROLLARY 16—Sharpness on an explicit neural continuation: *Let $\alpha \in \{1/2, 3/4\}$ and let n be a positive multiple of eight. For $K, R > 0$, put $c_n = R/n$ and*

$$v_n(z) = K\{\sigma(z - c_n) + \sigma(c_n - z)\} = K|z - c_n|.$$

This convex two-unit ReLU continuation satisfies

$$\rho_\alpha^{E_n}(v_n) - \rho_\alpha^{M_n}(v_n) = \frac{KR}{(1-\alpha)n^2}.$$

Consequently neither the exponent nor the leading constant in the noiseless width bound of Proposition 15 can be reduced uniformly over convex K-Lipschitz continuations, even on this explicit neural subclass. The conclusion concerns the stated midpoint and endpoint

rules, not a lower bound for all integration algorithms or a superiority claim for neural representations.

The same argument applies to a rectangular product innovation law when v is convex in each coordinate. If coordinate j has radius R_j, Lipschitz constant K_j and n_j bins, the product midpoint and product endpoint rules have width at most

$$\frac{1}{1-\alpha}\sum_j \frac{K_j R_j}{n_j^2}.$$

The number of evaluation points still grows as a product of the resolutions. This extension therefore retains the two-innovation cost rather than claiming that the curse of dimension has disappeared. It does not automatically apply to the possibly nonconvex minimum-of-cones representation of the general constructive backend. Convexity of a supplied continuation, or of the original value being enclosed, is a substantive hypothesis.

For the investment model, a midpoint affine support also supplies a feasible density witness for the all-original-action lower quadratic in Theorem 13. The new endpoint upper value and that lower quadratic therefore give

$$\underline{V}_0(x) \le V_0(x) \le J_0^{\widehat{\pi}}(x) \le \overline{J}_0(x).$$

The policy upper and lower costs use the same continuous innovation law. The state-cover residual remains a whole-domain statement, whereas an admission inequality is checked at the queried state. These two quantifiers are not interchangeable.

9.2. *Admission changes the statistical support*

A stored continuation first proposes an action. A qualified service accepts that action only when its complete-policy bound is at most τ against the original optimum. Otherwise it pays for the declared conventional action and verifies that action. An exact finite-cover fallback makes this a complete measurable service on the declared dyadic initial domain, rather than a controller defined only on successful sample rows. The fallback is part of the policy, and a rejected proposed action remains in the record.

THEOREM 17—Qualification and simultaneous resolving power: *Consider A complete services, fixed independently of an evaluation sample, each satisfying $0 \le J^a(x) - V^*(x) \le \tau$ for every state in the declared initial domain. Suppose each reported policy-cost enclosure has width at most w. For N independent initial states, let $[d_i^-, d_i^+]$ be the difference of two cost enclosures, intersected with $[-\tau, \tau]$. For any $r > 0$, all H declared pairwise intervals*

$$\left[\frac{1}{N}\sum_i d_i^- - r, \frac{1}{N}\sum_i d_i^+ + r\right] \tag{44}$$

cover their original-law mean contrasts with probability at least $1 - 2H\exp\{-Nr^2/(2\tau^2)\}$. Their half-widths are at most $r + w$, apart from the separately enclosed arithmetic used to report endpoints.

The global service definition is essential. Success on the observed states alone does not establish the support used in this theorem. Nor may a failed fitted proposal be deleted from a nominally unchanged-policy sample. The present service records its proposal, acceptance decision, final action, numerical refinement and exact-fallback entry. The original R95 policy experiment remains a different, unchanged experiment.

We use $\tau = 1/256$, $w = 1/8192$, $N = 4096$, $H = 12$ across two risk levels, and $r = 1/4096$. Then $2H\exp(-8) < 0.05$, while

$$r + w = \frac{3}{8192} < \frac{1}{2560} = \frac{\epsilon}{2}, \qquad \epsilon = \frac{1}{1280} = \frac{h(1/4)}{100}. \tag{45}$$

Thus the sample size and numerical precision are fixed together to resolve an average policy-cost difference at one percent of the largest first-date investment charge. The margin is eighty times smaller than the earlier $1/16$ margin. It is an explicit model-relative purchaser tolerance, not an estimated welfare weight or external monetary calibration. We report which comparisons actually meet it; the design alone implies no favorable sign.

9.3. *Acquisition, reuse and an economic purchase*

Let K_a be a one-time acquisition charge for service a, and c_a its cost per future workload. The acquisition account includes labels, feature search, fitting, validation, serialization and the chosen audit campaign. A fresh-process workload includes loading, identity checks, action proposal, admission, fallback, numerical certification, output and shutdown. Let $d = \mathbb{E}[J^a - J^b]$ be the original policy-cost contrast, $\lambda \geq 0$ the price of computational work in the same normalized cost units, and F an additional fixed fee difference. For m future workloads, the total normalized difference is

$$m\{d + \lambda(c_a - c_b)\} + \lambda(K_a - K_b) + F. \tag{46}$$

This expression distinguishes a policy-quality loss from the value of saved computation. A price is not inferred from seconds, a coefficient count, or an integration bin.

PROPOSITION 18—Robust acquisition-and-reuse region: *Suppose the policy contrast has simultaneous upper bound d^+, the workload saving $c_b - c_a$ has a valid lower bound s^-, and the incremental acquisition and fee have upper bounds k^+ and F^+. Then service a weakly improves total normalized cost whenever*

$$m(\lambda s^- - d^+) \geq \lambda k^+ + F^+. \tag{47}$$

If $\lambda s^- - d^+ > 0$ and the right side is nonnegative, the smallest sufficient integer workload count is its ratio rounded upward. If the denominator is nonpositive and the right side is positive, no positive workload count is certified by these bounds. Separately, a purchaser's quality condition is $d^+ \leq \epsilon$; it must not be inferred from a positive computational saving.

The timing repetitions below produce descriptive machine-specific ranges, not probabilistic guarantees for arbitrary future machines. Inserting their extrema in (47) gives a sensitivity calculation conditional on that timing envelope. It does not turn five repeated processes into a confidence bound on long-run prices or runtime. The theorem states the economic decision that adequately measured cost bounds

would license, while the experimental record states precisely which costs were observed.

10. A REMOTELY FROZEN, DECISION-SCALE COMPARISON

The new comparison follows the complete-service definition of Theorem 17. It is not a relabeling of the R95 actors. Oblique ReLU, adaptive coordinate-spline and projected-quadratic continuations have three nonlinear features and three affine coefficients. Each fitted design has rank six, uses 1,024 common original-risk training labels, and occupies an identical 128-byte executable encoding. Forward selection considers the same number of candidates and uses identical optimization stopping parameters. Every candidate fit, rank rejection and selected feature is retained. Effective design rank is reported separately from nonzero readout count and conditioning. Equal nominal or solver-call budgets do not imply equal fitting time.

The spline knots use training-state quantiles, and all feature selection uses training data only. Calibration uses 128 separate initial states at each risk level and fixes the selected service before confirmation. The source, six binary models, analysis rule and calibration selection are committed remotely before the new evaluation stream can open. Every arm remains in the comparison. This rank-six experiment repairs one specific confound in the preceding comparison; it is not an exhaustive frontier over ranks, labels, adaptive convex regression and computational budgets.

There are 4,096 new independent initial states per risk level, drawn from the explicitly specified uniform 20-bit dyadic distribution. Innovations between decisions remain continuous. All first proposals, rejected proposals, final actions, refinement levels and fallback flags are recorded. An accepted final action satisfies a $1/256$ original-optimum target and a $1/8192$ cost-width target. The comparison margin is $1/1280$, and the predeclared simultaneous half-width is strictly smaller than half that margin. None of these quantities may change after evaluation.

TABLE II

NEW-STREAM COMPLETE SERVICES AT THE ORIGINAL-OPTIMUM TARGET

Risk	Service	Mean cost midpoint	Mean numerical width	Maximum loss	Rejected proposals
0.50	Online risk-DP	0.6197369	0.0000011	0.0000017	0
0.50	Quadratic	0.6198883	0.0000011	0.0025887	0
0.50	ReLU	0.6199340	0.0000011	0.0039036	1136
0.50	Adaptive spline	0.6199407	0.0000011	0.0039033	1275
0.75	Online risk-DP	0.6308045	0.0000012	0.0000033	0
0.75	Quadratic	0.6309583	0.0000012	0.0029466	0
0.75	ReLU	0.6309673	0.0000012	0.0039032	1266
0.75	Adaptive spline	0.6310478	0.0000012	0.0039006	1073

Each row has 4,096 independent initial states, paired across services. Mean cost uncertainty here is numerical, not a sampling interval. Rejected first proposals receive the separately paid conventional fallback; their original actions and loss bounds remain in the raw data. The target distribution is the declared 20-bit dyadic initial law, with continuous innovations.

TABLE III: All prespecified paired mean-cost comparisons

Risk	Left minus right	Paired midpoint	Numerical half-width	Simultaneous interval
0.50	Online risk-DP minus Quadratic	-0.0001514	0.0000011	$[-0.0003967, 0.0000939]$
0.50	Online risk-DP minus ReLU	-0.0001971	0.0000011	$[-0.0004424, 0.0000481]$
0.50	Online risk-DP minus Adaptive spline	-0.0002038	0.0000011	$[-0.0004491, 0.0000414]$
0.50	Quadratic minus ReLU	-0.0000457	0.0000011	$[-0.0002910, 0.0001995]$
0.50	Quadratic minus Adaptive spline	-0.0000524	0.0000011	$[-0.0002977, 0.0001928]$
0.50	ReLU minus Adaptive spline	-0.0000067	0.0000011	$[-0.0002520, 0.0002386]$

Risk	Left minus right	Paired midpoint	Numerical half-width	Simultaneous interval
0.75	Online risk-DP minus Quadratic	-0.0001538	0.0000012	$[-0.0003991, 0.0000915]$
0.75	Online risk-DP minus ReLU	-0.0001628	0.0000012	$[-0.0004081, 0.0000825]$
0.75	Online risk-DP minus Adaptive spline	-0.0002433	0.0000012	$[-0.0004887, 0.0000020]$
0.75	Quadratic minus ReLU	-0.0000090	0.0000012	$[-0.0002543, 0.0002363]$
0.75	Quadratic minus Adaptive spline	-0.0000896	0.0000012	$[-0.0003349, 0.0001557]$
0.75	ReLU minus Adaptive spline	-0.0000805	0.0000012	$[-0.0003258, 0.0001648]$

The sampling radius is the predeclared $1/4096$ for every contrast; the family has coverage at least 95 percent. The exact comparison margin is $1/1280$, and the code rounds it inward when classifying a contrast.

The retained execution contains 32,768 arm–state records. Of the twelve fixed contrasts, 12 establish equivalence at the declared margin and 0 exclude zero. The largest simultaneous half-width is 0.00024531. There are 4750 rejected first proposals and 0 entries to the final exact fallback. These counts refer to the new complete services, not the historical R95 actors.

Construction and complete work. Label construction, all feature-search fits, state-cover qualification, calibration, serialization and reload are recorded separately for each representation. A model's standalone purchase bears the full shared label cost rather than an arbitrary one-third allocation. Five fresh processes per arm and risk level evaluate the same fixed 32-state timing workload; the parent timer includes interpreter startup, identity checking, loading, qualification, output and shutdown. CPU frequency is recorded, not controlled. Timing ranges are descriptive and feed a conditional sensitivity region, not a confidence interval for future hardware. One-

time independent audit cost is shown separately and can be included explicitly in the acquisition charge.

Prospective certificate ablation. The builders of the global norm, signed scalar and joint local-support bounds are fixed before confirmation, with the same residual cover and own-continuation resolution. Their proposed-policy identities remain explicit. A scalar bound for a rejected fitted proposal is not attached to the service's different fallback action. The comparison records qualification failures and paid construction work, while distinguishing these fixed builders from a separately tuned solver frontier. The earlier screening experiment and its negative net-time result remain unchanged.

Independent endpoints. The independent implementation uses exact rational arithmetic for all paired means and coverage constants. It does not import the production statistical formula. It also chooses eight random rows from each arm–risk stratum, reconstructs the stored candidate actions, and forms a finer original-law complete-policy enclosure and an original-action lower witness from separately written economic primitives. The finer interval must be contained in the reported one. Conventional proposal selection and the full fitting stack are not independently replayed. Sixty-four random endpoint checks are more informative than a fixed hand-selected fixture catalogue, but they are not an independent calculation of all 32,768 endpoints.

The experiment retains its precise state, control and horizon scope. The product quadrature result applies analytically to two innovation coordinates under its convexity hypotheses. It is not a completed two-control, two-innovation solver tournament, and the inherited adverse vector frontier remains visible. The revised comparison addresses the precision and identification questions within the original NBO program without borrowing evidence from an unexecuted model.

11. HORIZON COMPLEXITY AND THE PRICE OF ACCURACY

The inherited recursive query experiment exposes descendant growth. The own-future constructive backend has a different dependency graph: once f_{t+1} is stored, a query evaluates that representation and does not solve the remaining Bellman

problem again. This distinction changes the horizon account without removing any of the recursive benchmarks.

For each date let S_t be its state-context cover size, J_{ti} its feasible action-net size at node i, M_t its certified integration work in primitive representatives, and E_{t+1} the cost of evaluating its stored successor. Construction uses

$$Q_t = M_t \sum_{i \le S_t} J_{ti}, \qquad W \le \sum_{t<T} \{Q_t(C_t^{\text{prim}} + E_{t+1}) + C_t^{\text{compile}}\} + C_T^{\text{terminal}} + C^{\text{output}}. \tag{48}$$

The formula counts a directed acyclic sequence of stored continuations. A literal descendant solver instead multiplies successor solves by its action and shock branching factors. Neither expression is a lower bound for every possible algorithm. The parameters in (48) may themselves grow with horizon if the primitive regularity deteriorates.

PROPOSITION 19—Optimal dated error allocation: *Suppose a certified constructive implementation has a resource upper account $a_t \eta_t^{-p}$ for attaining total date-t residual allowance $\eta_t > 0$, with common $p > 0$. Each a_t includes action search, integration, feasibility, compilation and the cost of evaluating the stored successor; it cannot hide another recursive Bellman solve. Put $w_t = B_t > 0$ and*

$$S = \sum_t a_t^{1/(p+1)} w_t^{p/(p+1)}.$$

Among allocations satisfying $\sum_t w_t \eta_t \le \varepsilon$, the minimum of this resource upper model and its minimizer are

$$W^*_{\text{model}} = S^{p+1} \varepsilon^{-p}, \qquad \eta_t^* = \frac{\varepsilon}{S} (a_t / w_t)^{1/(p+1)}. \tag{49}$$

This optimizes the declared upper model, not the unknown fastest algorithm. Rounding each allowance down to a dyadic number within a factor two preserves the error constraint and increases that model by at most 2^p.

If $a_t \le A$ and $w_t = \beta^t$ for $0 < \beta < 1$, then

$$W^*_{\text{model}} \le A \varepsilon^{-p} (1 - \beta^{p/(p+1)})^{-(p+1)}.$$

With a maximum permissible allowance $\eta_{\max}$ and positive per-date setup costs, clipping the displayed allocation adds at most $\sum_t a_t \eta_{\max}^{-p}$ to this upper bound, plus the explicit setup account. For $w_t = 1$, the un-clipped bound is $A(T+1)^{p+1}\varepsilon^{-p}$. Thus uniform regularity and reusable stored continuations give polynomial, rather than descendant-tree, horizon dependence at fixed dimensions. They do not give dimension-free computation.

The power model follows, for example, when effective state, action and innovation covers and their certified errors have polynomial work bounds, successor evaluation has the separately proved compiled bound, and all moduli are uniformly controlled. The exponent must include all these factors. Tensor compilation in Theorem 3 has a 2^{p_t} query factor and cell-location work; a dense neural cone has an S_{t+1} factor. Those are different implementation accounts and must not be interchanged.

11.1. *Operation vectors, arithmetic and reliability*

We retain the resource vector

$$(N_{\text{primitive}}, N_{\text{descendant}}, N_{\text{interval}}, N_{\text{rational}}, N_{\text{network}}, B_{\text{read}}, B_{\text{write}}, M_{\text{peak}}).$$

Counts are reported before scalar prices are applied. Rational operations additionally record operand bit lengths; an arithmetic-operation count is not a bit-complexity bound. Multiplication and division costs at those lengths enter a bit-cost account. Loading, validation, serialization and shutdown belong to complete process time. Peak memory is a capacity requirement, not an additive flow cost unless its duration or rental convention is specified.

A fitting warning, invalid proposal, zero-progress probe, successful screen, protected fallback and terminal service status are different events. In particular, zero new fits cannot estimate fitting reliability, and a successful fallback cannot be counted as a successful prediction. The retained 336-service recursive catalogue and the nine-controller attained-cost experiment have different units of observation. New mathematical fixtures below establish the operation identities and allocation formula; they are not relabeled as a re-execution of the inherited eight-million-query vector tree or as a hardware calibration. The complete proof of Proposition 19 is in Supplement 10.

12. CONCLUSION

Neural Bellman Operators connect an attained own-future continuation to feasible decisions under the original economic law. The constructive neural realization, complete-policy comparison and explicit acquisition budget remain the paper's foundation. Under recursive risk, state-dependent qualification retains the original objective and the all-original-action lower obligation. The investment technology supplies primitive sufficient conditions under which these inequalities can actually be computed.

Convex-order risk integration adds a second-order precision account without smoothness of the risk-averse value. A complete admission-and-fallback service then turns the original-optimum guarantee into a support bound for paired economic inference. Together these results permit numerical resolution, sample size and a model-relative economic margin to be fixed before evaluation. This conclusion concerns the declared initial-state distribution and the original continuous innovation law, not an unreported change in either law.

The empirical questions remain distinct. A storage benefit need not be neural-specific; tighter uniform residuals need not yield better policy choices; and a valid screen need not save work after its construction is paid for. The new controlled-rank comparison and complete-service records make those distinctions observable. The acquisition-and-reuse region states which additional cost and price information supports an economic purchase. The fixed-accuracy horizon construction retains its explicit state, action, innovation and bit-growth costs; an unexecuted largest-vector comparison is not inferred from it.

The result is a constructive economic method with an original-law policy account and an explicit precision requirement. Its broader NBO applications, prior comparisons and adverse observations remain part of the preserved research record, under their own assumptions rather than an unsupported universal superiority assertion.

REFERENCES

ALOMARI, MOHAMMAD W. (2016): "A Generalization of Hermite–Hadamard's Inequality," arXiv:1603.08045. [4, 36]

FELZENSZWALB, PEDRO F. AND DANIEL P. HUTTENLOCHER (2012): "Distance Transforms of Sampled Functions," *Theory of Computing*, 8 (19), 415–428. [3, 8]

HOEFFDING, WASSILY (1963): "Probability Inequalities for Sums of Bounded Random Variables," *Journal of the American Statistical Association*, 58 (301), 13–30. [4]

MCSHANE, EDWARD J. (1934): "Extension of Range of Functions," *Bulletin of the American Mathematical Society*, 40 (12), 837–842. [3, 7]

MUNOS, REMI (2003): "Error Bounds for Approximate Policy Iteration," in *Proceedings of the Twentieth International Conference on Machine Learning*, 560–567. [3]

PIROTTA, MATTEO, MARCELLO RESTELLI, ALESSIO PECORINO, AND DANIELE CALANDRIELLO (2013): "Safe Policy Iteration," in *Proceedings of the 30th International Conference on Machine Learning*, vol. 28 of *Proceedings of Machine Learning Research*, 307–315. [3]

ROCKAFELLAR, R. TYRRELL AND STANISLAV URYASEV (2000): "Optimization of Conditional Value-at-Risk," *The Journal of Risk*, 2 (3), 21–41. [4, 36]

——— (2002): "Conditional Value-at-Risk for General Loss Distributions," *Journal of Banking and Finance*, 26 (7), 1443–1471. [21, 26]

RUSZCZYŃSKI, ANDRZEJ (2010): "Risk-Averse Dynamic Programming for Markov Decision Processes," *Mathematical Programming*, 125 (2), 235–261. [4, 23, 34]

——— (2014): "Erratum to: Risk-Averse Dynamic Programming for Markov Decision Processes," *Mathematical Programming*, 145, 601–604. [4, 23, 34]

RUSZCZYŃSKI, ANDRZEJ AND SHANGZHE YANG (2025): "Risk-Averse Control of Markov Systems with Value Function Learning," *Annals of Operations Research*. [4, 34]

SCHERRER, BRUNO, MOHAMMAD GHAVAMZADEH, VICTOR GABILLON, BORIS LESNER, AND MATTHIEU GEIST (2015): "Approximate Modified Policy Iteration and its Application to the Game of Tetris," *Journal of Machine Learning Research*, 16 (49), 1629–1676. [3]

YANG, INSOON (2020): "A Convex Optimization Approach to Dynamic Programming in Continuous State and Action Spaces," *Journal of Optimization Theory and Applications*, 187, 133–157. [3]